Supplementary Materials for:

MEMOL: Mixture of experts for multimodal learning through multi-head attention to predict drug toxicity

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Supplementary 1. Hyperparameter Settings.

Table S1. Hyperparameter configurations for MEMOL across different datasets.

Dataset	Learning rate	Batch Size	Encoder layers	Dropout ratio
hERG	0.001	256	1	0.0
DILI	0.0005	128	1	0.0
Skin Reaction	0.001	256	4	0.0
Carcinogenicity	0.001	256	4	0.3
DILIst	0.001	256	2	0.1

The selection of hyperparameters was based on systematic exploration to ensure stability and efficiency in training. We tested different learning rates, 0.001, 0.0005, and 0.0001, to evaluate their impact on convergence. The batch size was set to 128 or 256, balancing computational efficiency and training stability. To examine the effect of the number of MoE self-attention blocks, we varied it across 1, 2, 4, and 8 blocks. Furthermore, dropout regularization with values of 0.0, 0.1, and 0.3 was applied to prevent overfitting and improve generalization. The final hyperparameters were selected as summarized in Table S1.

Supplementary 2. Experiments under a Unified Data Splitting Strategy

Table S2. Comparative study with baseline models after unifying the train, validation, and test splits to an 8:1:1 ratio.

Dataset	hERG (8:1:1)		DILI (8:1:1)		Skin Reaction (8:1:1)	
Metric	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC
RF	0.8280	0.9130	0.8856	0.9083	0.6987	0.8056
LGBM	0.8156	0.9162	0.8769	0.8932	0.6400	0.7022
XGB	0.7778	0.8862	0.8885	0.9127	0.6587	0.7164
SVM	0.7825	0.8852	0.2019	0.4195	0.7013	0.7681
LR	0.6797	0.8146	0.7885	0.7887	0.4267	0.6289
ChemBERTa	0.8213	0.9214	0.8938	0.9313	0.7600	0.8562
MolCLR	0.8442	0.9176	<u>0.9359</u>	<u>0.9542</u>	0.7120	0.7971
GraphMVP	0.8128	0.9226	0.9241	0.9508	0.7440	0.8305
MolFormer	0.8430	0.9310	0.9121	0.9229	0.8000	0.8810
GIT-Mol	0.8527	<u>0.9365</u>	0.8883	0.9110	<u>0.8427</u>	<u>0.9095</u>
AMPred-CNN	0.7283	0.8730	0.8828	0.8761	0.7573	0.8774
CToxPred2	<u>0.8545</u>	0.9330	0.8901	0.9199	0.8187	0.8871
MEMOL	**0.8780**	**0.9400**	**0.9597**	**0.9734**	**0.8560**	**0.9229**

Across all three datasets, MEMOL demonstrated consistent improvements over the second-best models. On the hERG benchmark, it achieved an AUROC gain of 2.75%, along with a 0.37% increase in AUPRC. For DILI, the performance improvement was comparable, yielding 2.54% higher AUROC and 2.01% higher AUPRC. In the case of Skin Reaction, the AUROC margin was 1.58%, while the AUPRC improvement reached 1.47%, indicating that MEMOL provides balanced enhancements across both metrics. Under the 8:1:1 split setting, the results were consistent with those obtained using the 7:1:2 split described in the main text, confirming that MEMOL delivers superior performance regardless of the data partitioning strategy.

Supplementary 3. Statistical significance analysis

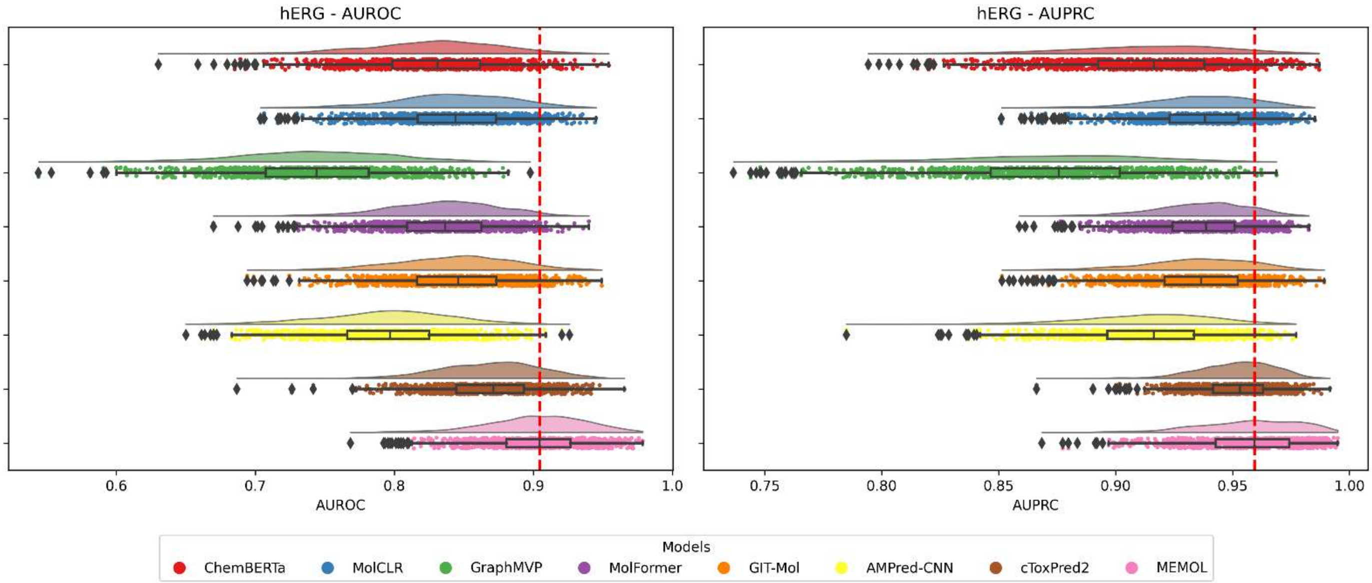

Figure S1. Raincloud plots of AUROC and AUPRC based on 1,000 bootstrap resamples for the proposed model and baseline models on the hERG dataset.

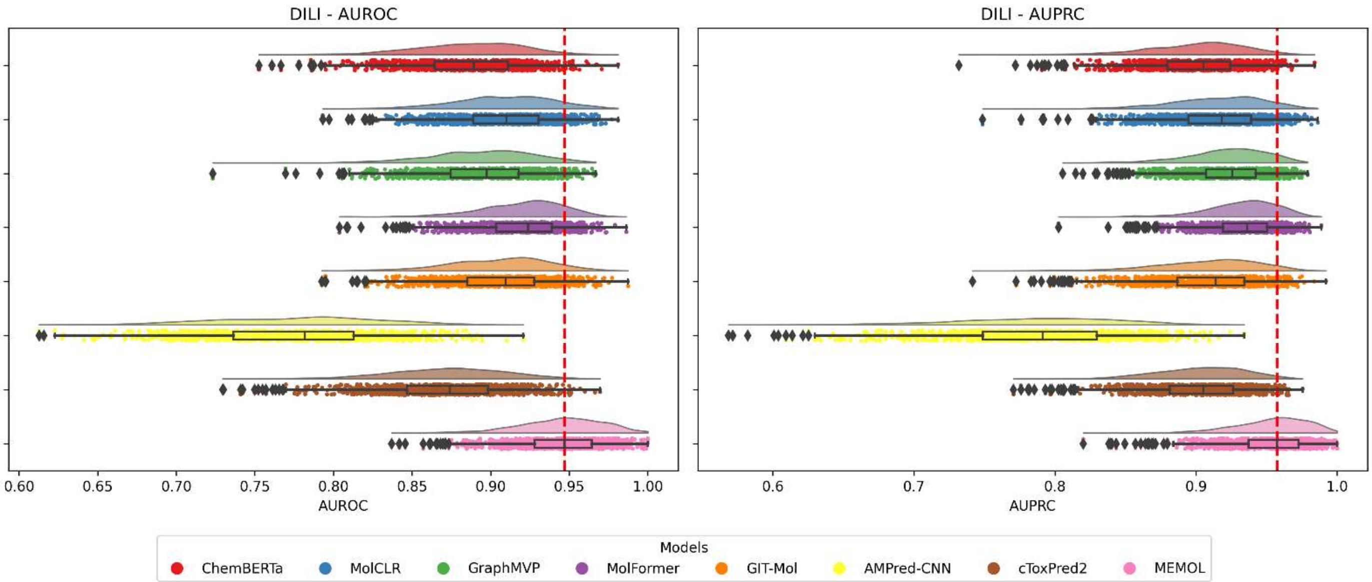

Figure S2. Raincloud plots of AUROC and AUPRC based on 1,000 bootstrap resamples for the proposed model and baseline models on the DILI dataset.

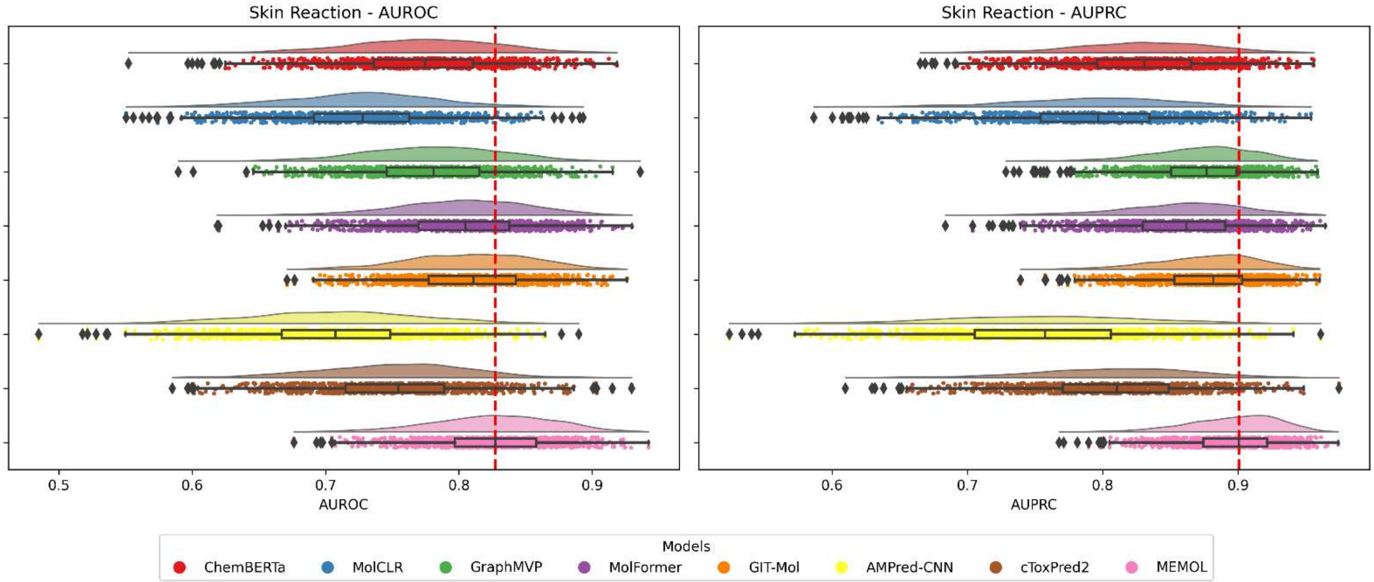

Figure S3. Raincloud plots of AUROC and AUPRC based on 1,000 bootstrap resamples for the proposed model and baseline models on the Skin Reaction dataset.

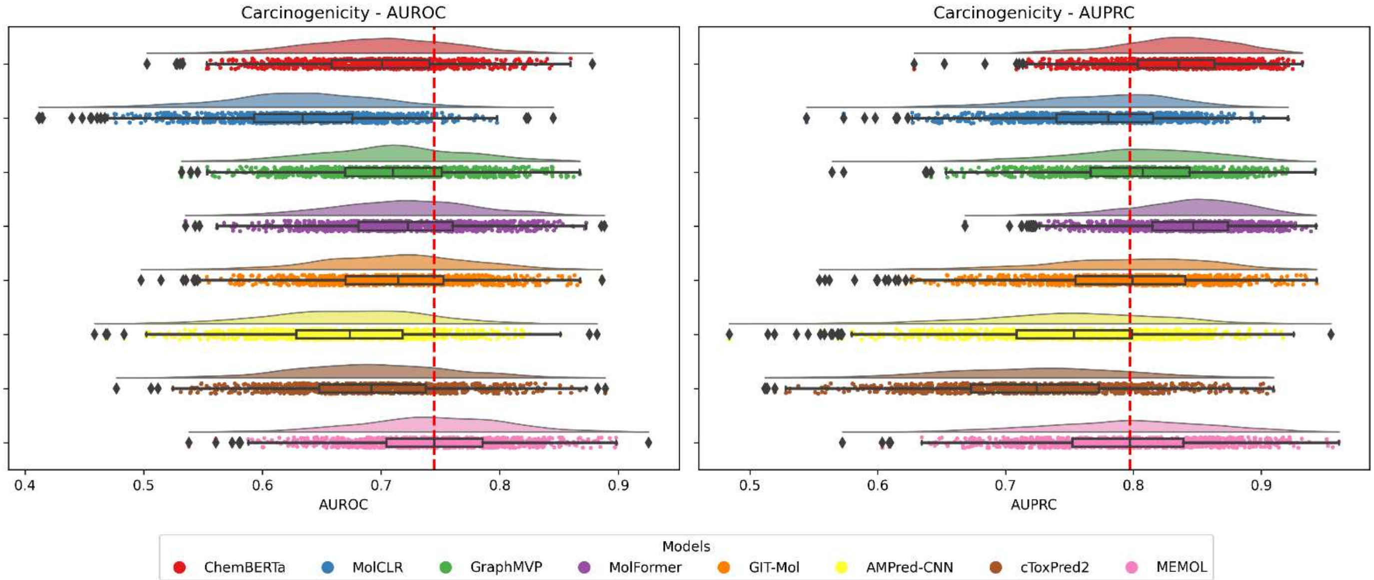

Figure S4. Raincloud plots of AUROC and AUPRC based on 1,000 bootstrap resamples for the proposed model and baseline models on the Carcinogenecity dataset.

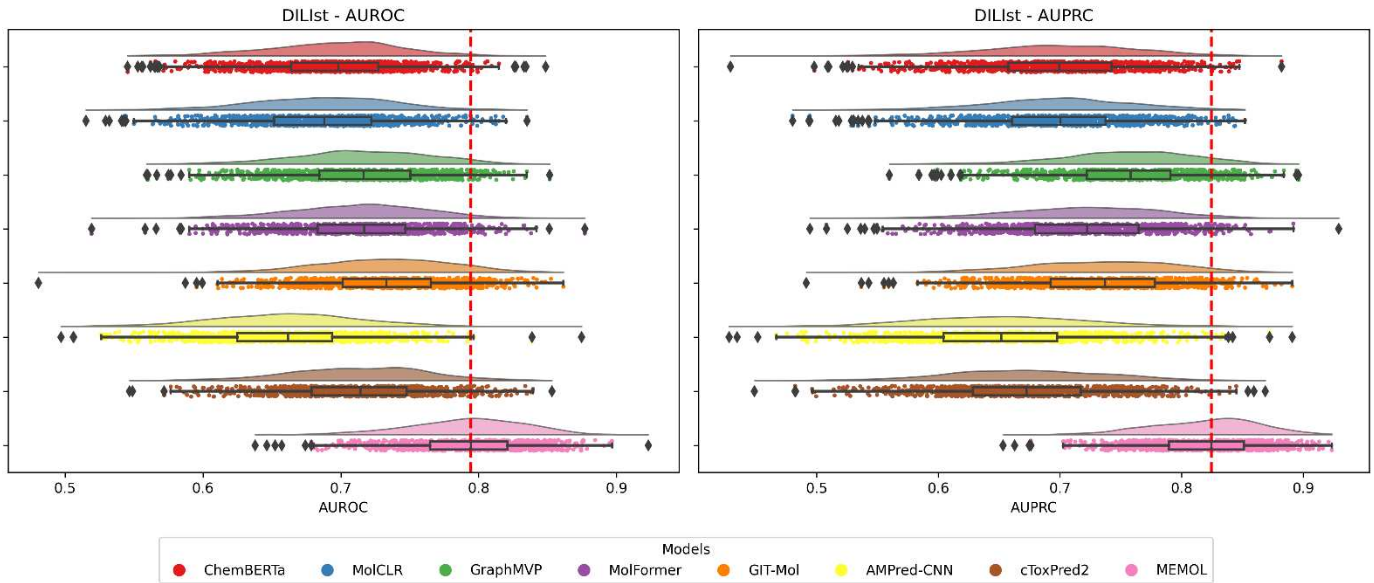

Figure S5. Raincloud plots of AUROC and AUPRC based on 1,000 bootstrap resamples for the proposed model and baseline models on the DILIst dataset.

To assess whether MEMOL's performance improvements were statistically significant, we employed a bootstrap resampling procedure. In this method, new test sets are repeatedly constructed by sampling with replacement from the original test set.

Each resampled set has the same size as the original, but some samples may appear multiple times while others may be omitted. This procedure simulates repeated experiments under slightly different data conditions. In our study, we generated 1,000 such bootstrap resamples for each dataset and recalculated AUROC and AUPRC for all competing deep learning models. The resulting distributions allowed us to estimate variance and construct confidence intervals, thereby providing a rigorous basis for significance testing.

Figure S1-S5 illustrates the distributions of AUROC and AUPRC obtained from 1,000 bootstrap resamples of the test sets. Overall, MEMOL consistently demonstrates performance distributions shifted to the right (i.e., toward higher accuracy) compared to baseline models. As shown in the boxplots, MEMOL also achieves the highest median values across all datasets, except for the Carcinogenicity dataset in terms of AUPRC. These results confirm that the superior performance of MEMOL is statistically significant and not attributable to random variation.

Supplementary 4. Model Interpretability by Attention Mechanisms

Figure S6. Attention score heatmap for the hERG dataset. Each column represents a SMILES sample, and each row corresponds to the relative contribution of a specific modality (Image, Graph, FP) under a particular attention head. Rows are indexed by head number (e.g., H1, H2, …), with modalities distinguished by their prefix (Image, Graph, FP). Thus, rows sharing the same head index indicate the modality-specific contributions within that head. The intensity reflects normalized weights in the range [0, 1], with darker shading indicating stronger contributions. Within each head, the three modality rows sum to one, directly showing the modality distribution learned by that head.

The fingerprint modality consistently receives the highest attention weights across all heads, as indicated by the darker shading of the fingerprint rows. This observation directly aligns with the main text experiments, where the fingerprint-only setting achieved the best performance among individual modalities. The heatmap thus confirms that the model effectively prioritizes fingerprint information when learning toxicity-related representations.

In the main text, the evaluation of modality combinations on hERG, the Image-only setting achieved slightly better performance than the Graph-only setting. A similar trend is visible in Figure S6: in Heads 2, 3, and 4, the Image rows appear to a small extent darker than the Graph rows. This consistency with the results reported in the main text indicates that the model captures the relative importance of images over graphs.

While fingerprints dominate overall, the distribution of weights is not uniform across attention heads. Some heads allocate nearly all weight to fingerprints, while others distribute more attention to images or graphs. This variation suggests that individual heads specialize in different aspects of molecular information, enabling the model to integrate complementary structural patterns. Such diversity across heads contributes to robustness and helps avoid bias toward a single modality.

Supplementary 5. Computational efficiency

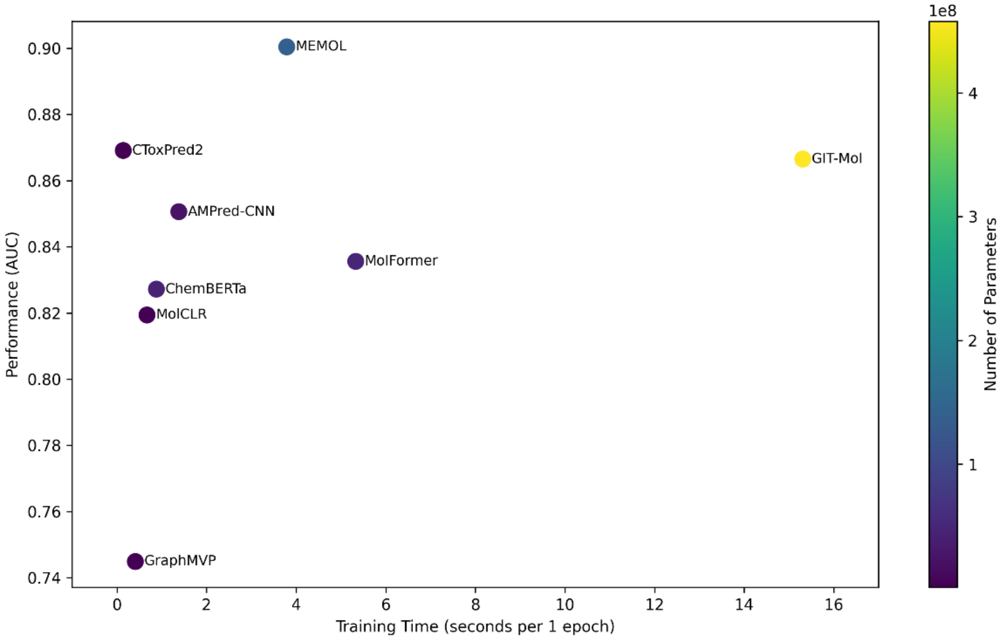

Figure S7. Comparison of model efficiency and performance on the hERG dataset. Each circle represents a model, where the vertical axis indicates predictive performance measured by AUROC, and the horizontal axis shows the training time per epoch (in seconds). The color scale corresponds to the number of model parameters, with larger parameter counts shown in brighter colors.

Several baseline models such as ChemBERTa, MolCLR, and GraphMVP are pretrained models, which only require fine-tuning on the downstream task. As a result, their training time per epoch is naturally much shorter, since the computationally expensive pretraining step is not included in our comparison. Similarly, CToxPred2 and AMPred-CNN achieve faster training speeds due to their relatively simple architectures, relying on a linear layer and a lightweight CNN, respectively.

On the other hand, GIT-Mol, as a multimodal learning architecture, involves substantial computation and requires significantly longer training time. In contrast, our proposed MEMOL, which also adopts a multimodal learning architecture, achieves a notable reduction in training time while maintaining high predictive preformance. Even compared with MolFormer, a recent transformer-based model, our approach shows a clear improvement in efficiency, training faster while still preserving high predictive accuracy.

Overall, although some models train faster than MEMOL, these models show lower predictive performance. MEMOL consistently delivers the highest AUROC among all compared methods, highlighting that our model achieves the best trade-off between efficiency and accuracy.